STUDENT RECORD SYSTEM

CODE COVERAGE REPORT

Statement and interpretation of executed source-line coverage

Document	TEST-COV-03
Project	Student Record System
Course	Software Engineering Mini Project
Team	Team 11
Implementation	C++17 application with SQLite persistence
Report basis	Latest successful GitHub Actions coverage run; reported line coverage confirmed as 76.34%.
Report date	11 November 2026

Prepared as a final testing report for the project documentation package.

1. Objective

This report presents source-line coverage from the latest successful GitHub Actions coverage run. Line coverage measures the proportion of instrumented executable source lines that were executed by the test workload.

2. Measurement Tools and Evidence

Item	Recorded value
Instrumentation	GCC coverage instrumentation (--coverage)
Report generator	GCOVR
Evidence file	coverage.xml generated by the GitHub Actions coverage job
Coverage package	src
Execution environment	Ubuntu GitHub Actions runner

3. Overall Line Coverage

76.34%	881 / 1,154	273
Line coverage	Lines executed / valid	Lines not executed

Calculation: 881 covered lines ÷ 1,154 valid lines × 100 = 76.343%, reported as 76.34%.

4. Interpretation

• 881 instrumented source lines were executed by the coverage workload.

• 273 valid source lines were not executed by that workload.

• The result reflects the source files included in the configured coverage report and exercised by the associated tests.

• A line not executed during testing is not necessarily defective; it identifies code paths not reached by the measured test workload.

5. Coverage Scope and Limitations

• The coverage package is the src directory selected by the GCOVR filter in the workflow.

• The report is based on the coverage.xml generated by the latest successful GitHub Actions coverage run.

• Coverage percentages depend on the source files included by the workflow's GCOVR filters and the tests executed in that run.

• Line coverage does not establish correctness, requirement completeness, or absence of defects.

• The aggregate line counts and percentage should be interpreted together; coverage alone does not measure the quality of assertions or the adequacy of test cases.

6. Result

The latest reported result is **76.34% line coverage**, corresponding to **881 of 1,154 valid lines** (273 lines not executed). This is the coverage figure confirmed for the latest successful run.

7. Conclusion

The coverage workload executed approximately 76.34% of the valid source lines included in the report. The remaining 23.66% were not executed by that workload and represent opportunities for additional test scenarios if greater line coverage is desired.

Coverage is a testing indicator, not proof that the application is defect-free. The project should continue to combine coverage measurement with functional, integration, security and performance testing.